1. IPv4 Addressing

Use 172.20.0.0/16 to create subnets that meet the host requirements.

- Staff: 90 hosts
- IT: 70 hosts
- Sales: 45 host

Subnet	1	2	4	8	16	32	64	128	256
Host	256	128	64	32	16	8	4	2	1
Subnet Mask	/24	/25	/26	/27	/28	/29	/30	/31	/32

- Staff: 90 = (___)/

IP start: _____._____._____._____

IP End:_____._____._____._____

- IT: 70 =(___)/

IP start: _____._____._____._____

IP End:_____._____._____._____

- Sales =(___)/

IP start: _____._____._____._____

IP End:_____._____._____._____

-Use 192.168.0.0/24 to create subnets that meet the host requirements

o Staff: 100 hosts

o Sales: 50 hosts

o IT: 25 hosts

Subnet	1	2	4	8	16	32	64	128	256
Host	256	128	64	32	16	8	4	2	1
Subnet Mask	/24	/25	/26	/27	/28	/29	/30	/31	/32

▪ Staff: 90 = (___)/

IP start: _____._____._____._____

IP End:_____._____._____._____

▪ IT: 70 =(___)/

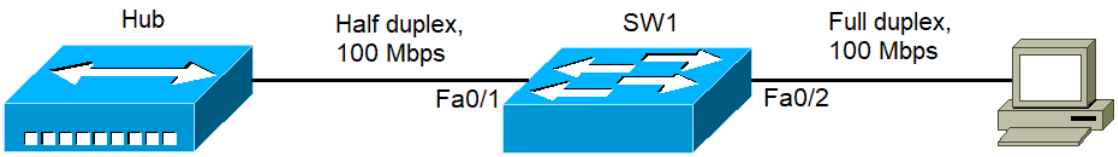
IP start: _____._____._____._____

IP End:_____._____._____._____

▪ Sales =(___)/

IP start: _____._____._____._____

IP End:_____._____._____._____



-Based on the attached image showing the network topology and interface settings, write the appropriate Cisco IOS commands to manually configure the speed and duplex settings for the indicated switch port

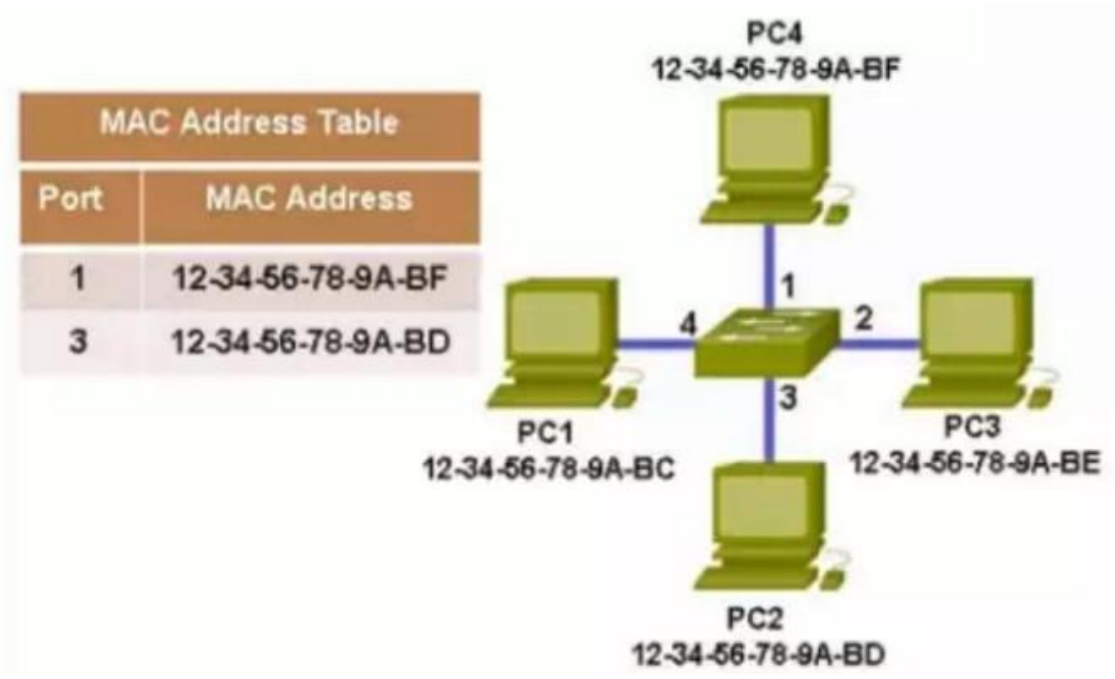
-Consider the following network. Suppose that PC1 formed a frame and sent it through the switch to be sent to PC2 Through which port(s) will the switch forward the frame?

a) 4 only

b) 3 only

c) 1,2,3 and 4

D) 1,2 and 3



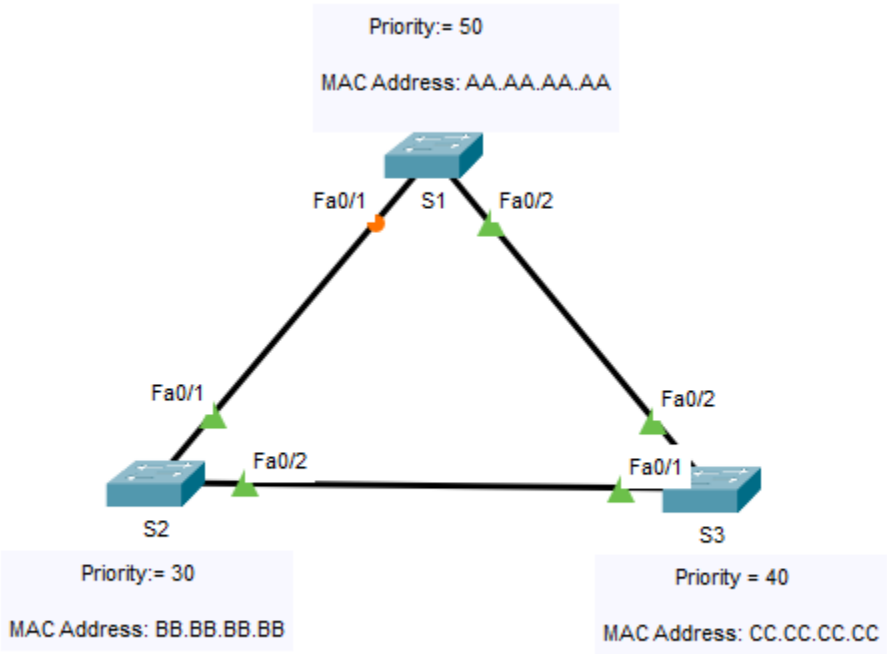
-Write down the commands on the router to accomplish the tasks given.

Question	Answer
Configure the device name on a switch as H12LAB	
Protect the access to the user mode using the password ER@#TYab	
Protect the remote access to the device using the password KG#@12ds	
Assign the last usable IP address in the subnet 192.168.15.96/29 to VLAN 1 interface of switch and enable it	
Assign the fifth usable address in the subnet 192.168.15.96/29 as the gateway for the switch	

-Create a new VLAN with the ID 3 and the name 'Oman'. Then, configure the FastEthernet 0/10 port to be assigned to this VLAN and set it to access mode. Write the necessary Cisco IOS commands.

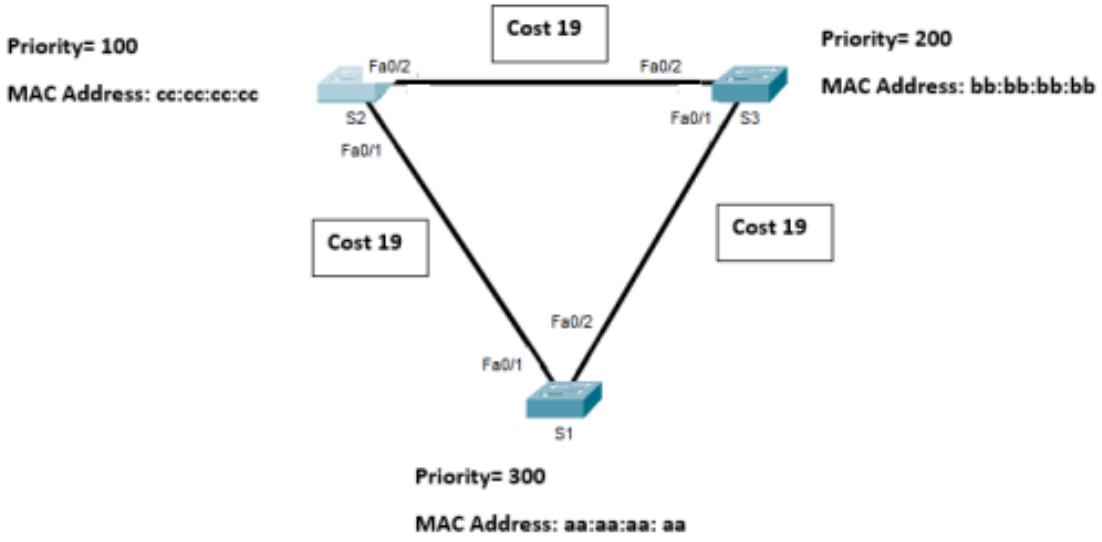
-Create VLAN 999 to be used as the native VLAN. Then, configure the FastEthernet 0/1 port to operate in trunk mode, and allow only the following VLANs on this trunk: 2, 500, 1001, and 999. Write the necessary Cisco IOS commands

Refer to the exhibit. Identify the Root Bridge, Designated .1 port (DP), Root port (RP) and Blocked port (BP) by filling the below table.



Switch Name Port Number	S1	Port Status	S2	Port Status	S3	Port Status
Fa0/1						
Fa0/2						
Root Bridge (Yes or No)						

Refer to the exhibit. Identify the Root Bridge, Designated port (DP), Root port (RP) and Blocked port (BP) by filling the below table.



Switch Name Port Number	S1	Port Status	S2	Port Status	S3	Port Status
Fa0/1						
Fa0/2						
Root Bridge (Yes or No)						

-Create an EtherChannel using Port-Channel 3 that includes interfaces FastEthernet 0/7 to FastEthernet 0/9. The channel should operate in active mode. Configure the interfaces to operate in trunk mode and allow only VLANs 2, 4, and 6 to pass through. Write the necessary Cisco IOS commands

-Write the appropriate Cisco IOS commands to enable DHCPv4 on the router with the following requirements:

1. Exclude the first 10 IP addresses from distribution.
2. The DHCP pool name should be **UTAS**.
3. The lease duration should be **7 days** only.
4. Configure the **domain name** and **DNS server** so that each client receives them automatically when assigned an IP address.

-Compare Stateless DHCPv6 and Stateful DHCPv6 using a table. Mention the advantages of each type and explain how they differ in assigning IPv6 addresses to devices.

Stateless DHCPv6	Stateful DHCPv6

-Compare **GUA** and **LLA**

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-Compare **RS** and **RA** messages

Explain the purpose of the following Port Security commands applied to a switch port, and how they help protect the network from threats:

```
Switch(config)# interface fa0/1
Switch(config-if)# switchport mode access
Switch(config-if)# switchport port-security
Switch(config-if)# switchport port-security maximum 1
Switch(config-if)# switchport port-security violation shutdown
Switch(config-if)# switchport port-security mac-address sticky
```

Compare the two attack types: Interception of Data and Man-in-the-Middle (MITM), in terms of method, level of risk, and the attacker's role in each. Provide examples if possible.

Explain how packets are forwarded in Fast Switching, and highlight the key differences between Fast Switching and CEF (Cisco Express Forwarding)